Naive Bayes: Bernoulli and Multinomial MLE

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Abstract

This short note shows that the maximum-likelihood estimators for Bernoulli and Multinomial Naive Bayes admit closed-form solutions in both cases.

1 Introduction

Let

$$\mathcal{D} = \{(x^{(r)}, y_r)\}_{r=1}^{n}$$

be a labeled training set, and let

$$V = \{w_1, \ldots, w_d\}$$

be the vocabulary, e.g. the set of all words in the training set. Finally consider,

$$\mathcal{Y} = \{y_1, \ldots y_K\}$$

to be the set of all possible classes/labels.

For a fixed class k, define

$$A_k := \{r : y_r = k\}, \qquad N_k := |A_k|.$$

The Bernoulli and Multinomial versions of Naive Bayes correspond to two different probabilistic descriptions of a document.

The aim below is to derive the maximum-likelihood estimators of these two class-conditional probabilities, which serves as our machine learning *objective*. Interestingly, in both cases the optimization admits a *closed-form solution*, so fitting the model reduces to computing suitable empirical counts from the training set.

In *Bernoulli NB*, each coordinate records only whether a vocabulary word appears

$$X_j^{(r)} \in \{0, 1\}.$$

Thus the natural class-conditional quantity to estimate is

$$X_j \mid Y = k \sim \text{Bernoulli}(\theta_{k,j}), \qquad \theta_{k,j} = P(X_j = 1 \mid Y = k),$$

namely the probability that a class-k document contains the word w_j.

In *Multinomial NB*, instead,

$$X_j^{(r)} \in \mathbb{N}_0$$

records the number of occurrences of w_j. Conditional on the class, the tokens of a document are modeled as i.i.d. categorical draws from a class-specific distribution over the vocabulary. The corresponding quantity

$$W \mid Y = k \sim \text{Categorical}(\theta_{k,1}, \ldots, \theta_{k,d}), \qquad \theta_{k,j} = P(W = w_j \mid Y = k).$$

Then the count vector

$$X^{(r)} \mid Y = k \sim \text{Multinomial}(m_r; \theta_{k,1}, \ldots, \theta_{k,d}).$$

We stress that, despite using the same notation, the parameter $\theta_{k,j}$ has a different probabilistic meaning in the two models; in Bernoulli NB it is a document-level occurrence probability, whereas in Multinomial NB it is a token-level probability.

At the dataset level, the labeled documents are assumed to be *i.i.d.* In the Bernoulli model, the word-presence indicators are additionally assumed *conditionally independent given the class.* In the Multinomial model, the conditional independence assumption is instead imposed at the token level; the resulting word-count vector has a multinomial distribution.

Under these assumptions, the maximum-likelihood problem can be treated separately for each class k. Thus, for both models, it is enough to fix a class k and optimize the corresponding parameter vector $\theta_k = (\theta_{k,1}, \ldots, \theta_{k,d})$. For the Bernoulli NB in particular, the conditional independence allows us to estimate each $\theta_{k,j}$ separately too. See Section A for more details.

2 Bernoulli Naive Bayes

For Bernoulli NB,

$$x_j^{(r)} = \begin{cases} 1, & \text{if word } w_j \text{ occurs in document } r, \\ 0, & \text{otherwise.} \end{cases}$$

For each class k and word w_j, let

$$\theta_{k,j} = P(X_j = 1 \mid Y = k).$$

Thus

$$X_j \mid Y = k \sim \text{Bernoulli}(\theta_{k,j}).$$

For fixed k, j, the likelihood is

$$L(\theta_{k,j}) = \prod_{r \in A_k} \theta_{k,j}^{x_j^{(r)}} (1 - \theta_{k,j})^{1 - x_j^{(r)}}.$$

Hence

$$\ell(\theta_{k,j}) = \sum_{r \in A_k} \left[x_j^{(r)} \log \theta_{k,j} + (1 - x_j^{(r)}) \log(1 - \theta_{k,j}) \right].$$

Differentiating and setting the derivative equal to zero gives

$$\frac{\sum_{r \in A_k} x_j^{(r)}}{\theta_{k,j}} - \frac{N_k - \sum_{r \in A_k} x_j^{(r)}}{1 - \theta_{k,j}} = 0,$$

where $N_k; = |A_k|$ denotes the number of all training documents belonging to class k.

Therefore,

$$\boxed{\theta_{k,j}^{*,B} = \frac{\sum_{r \in A_k} x_j^{(r)}}{N_k}.}$$

Thus

$$\boxed{\theta_{k,j}^{*,B} = \frac{\#\{\text{class-}k \text{ documents containing } w_j\}}{\#\{\text{class-}k \text{ documents}\}}.}$$

3 Multinomial Naive Bayes

For Multinomial NB,

$$x_j^{(r)} = \text{number of occurrences of } w_j \text{ in document } r.$$

For each class k, let

$$\theta_k = (\theta_{k,1}, \ldots, \theta_{k,d}), \qquad \sum_{j=1}^{d} \theta_{k,j} = 1,$$

where

$$\theta_{k,j} = P(W = w_j \mid Y = k).$$

If document r has

$$m_r = \sum_{j=1}^{d} x_j^{(r)}$$

tokens, then

$$X^{(r)} \mid Y = k \sim \text{Multinomial}(m_r; \theta_{k,1}, \ldots, \theta_{k,d}).$$

Hence, up to factors independent of θ_k,

$$L(\theta_k) \propto \prod_{r \in A_k} \prod_{j=1}^{d} \theta_{k,j}^{x_j^{(r)}}.$$

Therefore,

$$\ell(\theta_k) = \sum_{j=1}^{d} \left(\sum_{r \in A_k} x_j^{(r)} \right) \log \theta_{k,j} + C.$$

We maximize this subject to

$$\sum_{j=1}^{d} \theta_{k,j} = 1.$$

Using a Lagrange multiplier,

$$\mathcal{L} = \sum_{j=1}^{d} \left(\sum_{r \in A_k} x_j^{(r)} \right) \log \theta_{k,j} - \lambda \left(\sum_{j=1}^{d} \theta_{k,j} - 1 \right).$$

The first-order conditions give

$$\theta_{k,j} = \frac{\sum_{r \in A_k} x_j^{(r)}}{\lambda}.$$

The condition

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0$$

gives

$$\sum_{j=1}^{d} \theta_{k,j} = 1,$$

d therefore

$$\lambda = \sum_{r \in A_k} \sum_{\ell=1}^{d} x_\ell^{(r)}.$$

Hence

$$\boxed{\theta_{k,j}^{*,M} = \frac{\sum_{r \in A_k} x_j^{(r)}}{\sum_{r \in A_k} \sum_{\ell=1}^{d} x_\ell^{(r)}}.}$$

Thus

$$\boxed{\theta_{k,j}^{*,M} = \frac{\#\{\text{occurrences of } w_j \text{ in class } k\}}{\#\{\text{all token occurrences in class } k\}}.}$$

A Independence and MLE

These assumptions also make the maximum-likelihood problem separable across classes. If $\theta = (\theta_1, \ldots, \theta_K)$, with θ_k denoting the parameters of the class-k conditional model, then the class-conditional part of the log-likelihood can be written as

$$\ell(\theta; \mathcal{D}) = \sum_{k=1}^{K} \sum_{r \in A_k} \log p(x^{(r)} \mid Y = k; \theta_k).$$

This follows from the i.i.d. assumption on the training set, which allows the likelihood to factorize over observations, together with the fact that the class label of each training observation is known.

Since the term corresponding to class k depends only on θ_k

$$\boxed{\arg\max_{\theta} \ell(\theta; \mathcal{D}) = \left(\arg\max_{\theta_1} \ell_1(\theta_1), \ldots, \arg\max_{\theta_K} \ell_K(\theta_K) \right),}$$

where

$$\ell_k(\theta_k) = \sum_{r \in A_k} \log p(x^{(r)} \mid Y = k; \theta_k).$$

Thus the likelihood parameters can be estimated *independently* for each class.

For Bernoulli NB, *conditional independence* gives the further factorization

$$p(x^{(r)} \mid Y = c_k; \theta_k) = \prod_{j=1}^{d} p(x_j^{(r)} \mid Y = c_k; \theta_{k,j}),$$

so the optimization also separates across vocabulary words. Hence each $\theta_{k,j}$ can be estimated independently.

For Multinomial NB, the word counts are *not* independent, since

$$\sum_{j=1}^{d} x_j^{(r)} = m_r.$$

However, the multinomial log-likelihood has the form

$$\ell_k(\theta_k) = C + \sum_{j=1}^{d} \left(\sum_{r \in A_k} x_j^{(r)} \right) \log \theta_{k,j},$$

subject to

$$\sum_{j=1}^{d} \theta_{k,j} = 1.$$

Thus we again obtain one parameter $\theta_{k,j}$ for each class and each word, but the parameters within a class are optimized jointly under the normalization constraint.

We therefore can focus on deriving the closed-form estimators

$$\theta_{k,j}^{*}.$$